

Topics Covered:

- Notation for Frames, Points, and Rotations
- The Three Uses of a Rotation Matrix
- Deriving the Rotation Matrix (Change of Basis)
- Rotation Matrices as a Group: $SO(2)$
- Multiple Displacements

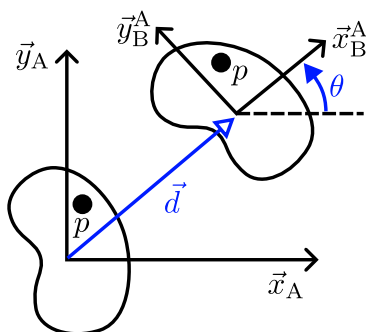
Additional Reading:

- Lynch, K.M. and Park, F.C. Modern Robotics: Section 3.2
- Craig, J.J. Introduction to Robotics: 2.3
- Murray et al. A Mathematical Introduction to Robotic Manipulation: Chapter 2, Section 2.1
- 3Blue1Brown Video on Change of Basis: <https://www.youtube.com/watch?v=P2LTAU01TdA>

Review

Last class we went over how to describe the configuration of a rigid body. We determined that we need 3 degrees of freedom to describe planar transformations: x , y , and θ .

We ended the last class with the 3DOF example:



Configuration of a rigid body:

$$g = (x, y, \theta)^\top \quad (\text{Vector Notation})$$

or equivalently:

$$g = (x, y, R(\theta))$$

$$g = (\vec{d}, R(\theta))$$

$$g = (\vec{d}, \theta)$$

$$\text{Note: } R(\theta) := \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

In today's class, we will go over how to represent both configurations and transformations using rotation matrices.

Notation

The General Rule

Subscripts say which frames are involved. Superscripts say which frame you are measuring in.

Read $\vec{d}_{AB}^A$ as: “the displacement from A to B, written down by an observer sitting in frame A.”

For the sake of consistent notation moving forward, we will define and use the following:

- $\hat{x}_A, \hat{y}_A$: the unit vectors along the x and y directions of frame A
- $\hat{x}_B^A$: the x axis of frame B , expressed in the coordinates of frame A
- g_{AB} : the transformation from frame A to frame B
- $\vec{d}_{AB}^A$: the displacement from frame A to frame B , in reference to frame A
- θ_{AB} : the angle from frame A to frame B
- $R(\theta_{AB})$: the rotation matrix built from the angle θ_{AB}
- $\vec{p}_B^A$: a point p attached to frame B , expressed in frame A
- $\vec{v}^A$: general vector notation, expressed in the coordinates of frame A

The Three Uses of a Rotation Matrix

A rotation matrix can be interpreted as being used for three different purposes.

1. **To represent an orientation.** $R(\theta_{AB})$ is the orientation of frame B with respect to frame A .
2. **To rotate a vector or a frame** (an **active** rotation). Given a point $\vec{p}$, the point

$$\vec{q} = R(\theta) \vec{p}$$

is a *different* point, obtained by physically spinning $\vec{p}$ about the origin by θ . Both $\vec{p}$ and $\vec{q}$ are measured in the *same* frame.

3. **To change the reference frame** in which a vector is expressed (a **passive** rotation). Given one fixed vector $\vec{v}$ whose coordinates in frame A are known, and frames A and B sharing an origin,

$$\vec{v}^B = R(\theta_{AB})^{-1} \vec{v}^A \quad \Leftrightarrow \quad \vec{v}^A = R(\theta_{AB}) \vec{v}^B$$

gives the coordinates of that *same* vector as measured in frame B .

Note the shared origin. Rotation matrices alone cannot account for the displacement $\vec{d}_{AB}^A$ between the two origins; we return to this later.

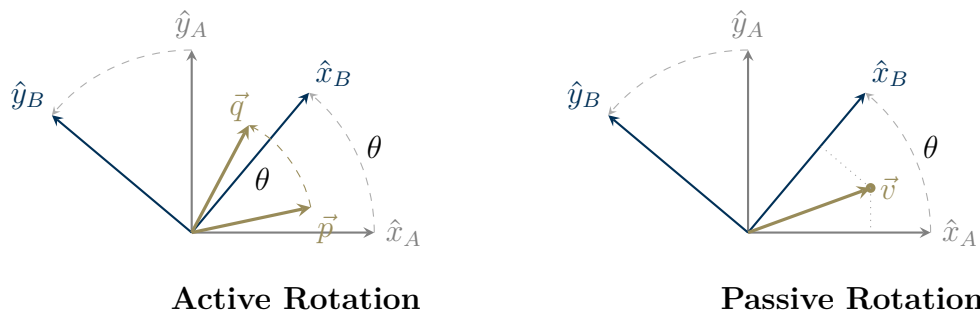


Figure 1: The same matrix $R(\theta)$ answers both questions, but the questions are different. On the left, $\vec{p}$ is carried to a new location $\vec{q}$. On the right, the single vector $\vec{v}$ is given two sets of coordinates, $\vec{v}^A$ and $\vec{v}^B$.

Deriving the Rotation Matrix

Consider the following diagram:

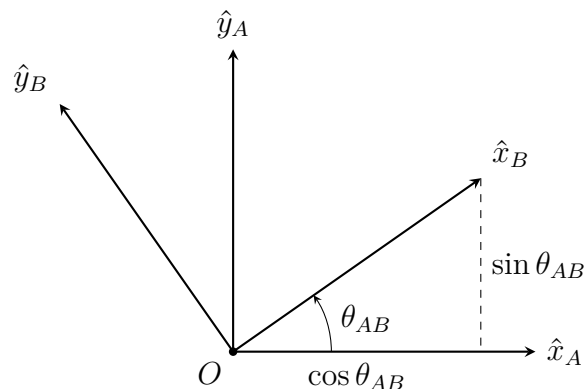


Figure 2: The rotated frame $\{B\}$ expressed relative to frame $\{A\}$. The columns of the rotation matrix are the unit axes of $\{B\}$ expressed in coordinates of $\{A\}$.

Reading the axes of frame B off the diagram in A 's coordinates:

$$\hat{x}_B^A = \begin{bmatrix} \cos(\theta_{AB}) \\ \sin(\theta_{AB}) \end{bmatrix}, \quad \hat{y}_B^A = \begin{bmatrix} -\sin(\theta_{AB}) \\ \cos(\theta_{AB}) \end{bmatrix}$$

Stacking these two vectors side by side as columns gives our canonical rotation matrix:

$$R(\theta_{AB}) = \begin{bmatrix} | & | \\ \hat{x}_B^A & \hat{y}_B^A \\ | & | \end{bmatrix} = \begin{bmatrix} \cos(\theta_{AB}) & -\sin(\theta_{AB}) \\ \sin(\theta_{AB}) & \cos(\theta_{AB}) \end{bmatrix}$$

Note that we can translate between a matrix representation of a rotation and the corresponding rotation angle using:

$$R = \begin{bmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} \end{bmatrix} \quad \theta = \text{atan2}(R_{21}, R_{11})$$

This derivation can be connected back to linear algebra.

Linear Algebra Connection: Change of Basis

Youtube video on change of basis: <https://www.youtube.com/watch?v=P2LTAU01TdA>
 A vector is a geometric object independent of the basis used to describe it, but its coordinates depend on that basis. If $P_{A \rightarrow B}$ is the change-of-basis matrix from basis A to basis B , then

$$\underbrace{\vec{v}^A}_{\text{coord. in A}} = P_{A \rightarrow B} \underbrace{\vec{v}^B}_{\text{coord. in B}}$$

where $\vec{v}$ is the same vector, expressed in different frames.

To connect the rotation matrix back to our uses mentioned earlier, consider the following example.

Example: Take a vector $\vec{v}$ whose coordinates in frame A are

$$\vec{v}^A = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \quad \theta = 30^\circ, \quad R(30^\circ) = \begin{bmatrix} 0.866 & -0.500 \\ 0.500 & 0.866 \end{bmatrix}$$

There are two entirely different things we can do with $R(30^\circ)$ here, and they give different answers.

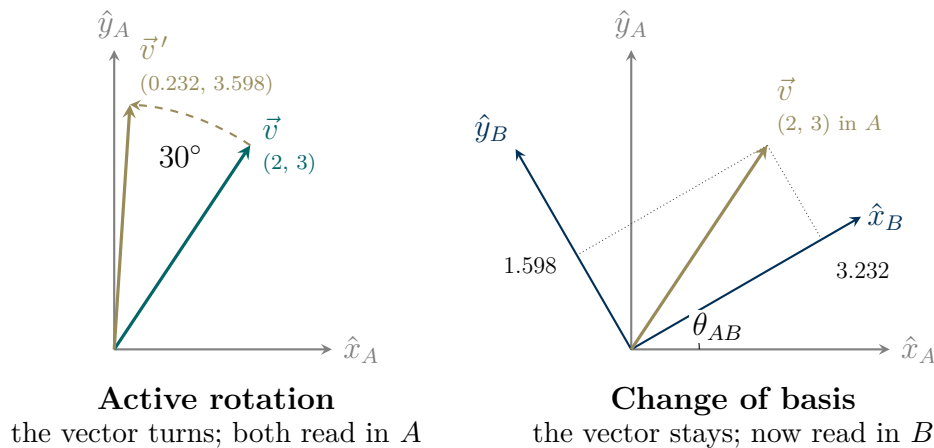


Figure 3: One starting vector, two different operations. On the left the vector moves and frame A is the only frame in sight. On the right nothing moves; we simply resolve the same arrow along the axes of a second frame B .

Option 1: active rotation. Spin the vector through 30° and keep measuring in frame A :

$$\vec{v}'^A = R(30^\circ) \vec{v}^A = \begin{bmatrix} 0.866 & -0.500 \\ 0.500 & 0.866 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1.732 - 1.500 \\ 1.000 + 2.598 \end{bmatrix} = \begin{bmatrix} 0.232 \\ 3.598 \end{bmatrix}$$

This is a *new* vector, written in the *same* frame.

Option 2: change of basis. Leave the vector exactly where it is and re-express it in frame B , where B is frame A rotated by $\theta_{AB} = 30^\circ$. Our change of basis formula runs $B \rightarrow A$, so to go $A \rightarrow B$ we need its inverse:

$$\vec{v}^B = R(\theta_{AB})^{-1} \vec{v}^A = R(\theta_{AB})^\top \vec{v}^A = \begin{bmatrix} 0.866 & 0.500 \\ -0.500 & 0.866 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1.732 + 1.500 \\ -1.000 + 2.598 \end{bmatrix} = \begin{bmatrix} 3.232 \\ 1.598 \end{bmatrix}$$

This is the *same* vector, written in a *different* frame.

Rotation Matrices as a Group: $SO(2)$

A rotation matrix is a special type of matrix that encodes a rotation in space, transforming vectors without changing their length or the angles between them. These matrices form a *group* under multiplication, meaning that combining rotations, doing nothing, or undoing a rotation all result in another valid rotation matrix. We will unpack this statement in this section.

Overall, rotation matrices have two important properties:

- $R^T R = I$ (unit axes are orthogonal)
- $\det R = 1$ (rotations follow the right-hand rule, i.e. $\vec{x} \times \vec{y} = \vec{z}$)

The first property stems from two conditions placed on our derivation of the rotation matrix: a unit norm condition specifying that the coordinate axes are represented by unit vectors, and an orthogonality condition on the unit axes. Together, these four constraints (2 for the unit norm axes and 2 for the orthogonality conditions) can be represented by the single constraint $R^T R = I$.

The second property is a result of the fact that the rotation matrix is right handed (i.e., $\vec{x} \times \vec{y} = \vec{z}$).

Due to these properties, rotation matrices can be mathematically classified as a *special orthogonal group* $SO(2)$. This is called a “group” because it satisfies the properties required of a mathematical group.

Definition: Group

A group is a set of elements $G = \{a, b, c, \dots\}$ with a binary operation $\cdot$ that satisfies the following properties:

closure	$a \cdot b \in G$ for all $a, b \in G$
associativity	$(a \cdot b) \cdot c = a \cdot (b \cdot c)$
identity element exists	there is an $I \in G$ such that $a \cdot I = I \cdot a = a$ for each $a \in G$
inverse exists	for each $a \in G$ there exists $a^{-1} \in G$ such that $a \cdot a^{-1} = a^{-1} \cdot a = I$

For the special orthogonal group $SO(2)$, the group operation is matrix multiplication, and the following group properties are:

closure	$R_1 R_2 \in SO(2)$
associativity	$(R_1 R_2) R_3 = R_1 (R_2 R_3)$
identity element exists	$I \in SO(2)$
inverse exists	matrix inverse $R^{-1} = R^T$

Proofs of these properties are as follows.

Properties of the special orthogonal group $SO(2)$

Proposition 1. *The inverse of a rotation matrix $R \in SO(2)$ is also a rotation matrix, and it is equal to the transpose of R , i.e., $R^{-1} = R^T$.*

Proof. The condition $R^T R = I$ implies that $R^T = R^{-1}$ and $RR^T = I$. Since $\det R^T = \det R = 1$, R^T is also a rotation matrix. ■

Proposition 2. *The product of two rotation matrices is a rotation matrix*

Proof. Given $R_1, R_2 \in SO(2)$, their product $R_1 R_2$ satisfies $(R_1 R_2)^T R_1 R_2 = R_2^T R_1^T R_1 R_2 = R_2^T R_2 = I$. Also, $\det R_1 R_2 = \det R_1 \cdot \det R_2 = 1$. Thus, $R_1 R_2$ satisfies both conditions for a rotation matrix. ■

Proposition 3. *Multiplication of rotation matrices is associative, i.e., for $R_1, R_2, R_3 \in SO(2)$, $(R_1 R_2) R_3 = R_1 (R_2 R_3)$.*

Proof. Matrix multiplications are associative. ■

Proposition 4. *For any vector $x \in \mathbb{R}^2$ and $R \in SO(2)$, the vector $y = Rx$ has the same length as x .*

Proof. This follows from:

$$\|y\|^2 = y^T y = (Rx)^T Rx = x^T R^T Rx = x^T x = \|x\|^2$$

Later in the course we will see that these same properties hold for the special group of rotation matrices for three-dimensional transformations, called the *special orthogonal group* $SO(3)$. The only property that differs for rotation matrices in higher dimensions is that the rotation matrices are not generally commutative (i.e., $R_1 R_2 \neq R_2 R_1$).

Multiple Displacements

So far we have only rotated about a shared origin. Adding a displacement, the transformation of a point from frame B into frame A is

$$\vec{p}_B^A = \vec{d}_{AB}^A + R(\theta_{AB}) \vec{p}^B$$

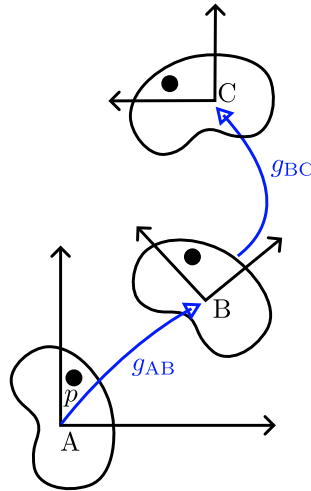
which is use 3 from earlier, with an offset added on. Note that the pair $(\vec{d}, R(\theta))$ can describe either a **configuration** or a **transformation**:

- If $\vec{d}_{BC} = 0$, then the transformation is a pure rotation.

- If $\theta_{BC} = 0$, i.e. $R(\theta_{BC}) = \mathbb{1}$, then the transformation is a pure translation.

Now, what about multiple displacements?

Consider the following example:



So far, we have only considered the transformation of a point from one frame to another. For example, in this example:

$$p_C \text{ in frame B is } p_C^B = \vec{d}_{BC}^B + R(\theta_{BC})\vec{p}$$

$$p_B \text{ in frame A is } p_B^A = \vec{d}_{AB}^A + R(\theta_{AB})\vec{p}$$

But what if we want to find p_C in frame A? We can use the following relationship:

$$\begin{aligned} p_C^A &= \vec{d}_{AB}^A + R(\theta_{AB})p_C^B \\ &= \vec{d}_{AB}^A + R(\theta_{AB}) \left(\vec{d}_{BC}^B + R(\theta_{BC})\vec{p} \right) \\ &= \vec{d}_{AB}^A + R(\theta_{AB})\vec{d}_{BC}^B + R(\theta_{AB})R(\theta_{BC})\vec{p} \end{aligned}$$

We can write this multiple transformation in vector coordinate form as:

$$\begin{aligned} g_{AC} &= g_{AB} \cdot g_{BC} \\ &= (\vec{d}_{AB}^A, R(\theta_{AB})) \cdot (\vec{d}_{BC}^B, R(\theta_{BC})) \\ &= \underbrace{(\vec{d}_{AB}^A + R(\theta_{AB})\vec{d}_{BC}^B)}_{\vec{d}_{AC}^A}, \underbrace{R(\theta_{AB})R(\theta_{BC})}_{R(\theta_{AC})} \end{aligned}$$

Notice that the subscripts chain the way the notation rule promises: an AB followed by a BC collapses to an AC . Next class we will formalize this composition as a *product*, and derive the inverse transformation.